

Proposed Solution – NHAID SecureID: Offline Facial Authentication System for Remote Operations

Our proposed solution, **NHAID SecureID**, is a lightweight, AI-powered, fully offline facial recognition and liveness detection system specifically designed for field personnel authentication in remote and zero-network environments. The solution addresses the core challenge of securely verifying the identity of workers on National Highway project sites without relying on internet connectivity while maintaining high accuracy, fast processing speed, and seamless integration with the existing Datalake 3.0 ecosystem.

The system follows a multi-layered Edge AI architecture optimized for deployment on standard mid-range Android and iOS devices. The authentication pipeline begins with **MediaPipe Face Detection**, which rapidly detects and localizes faces in real time. The detected face then passes through **MiniFASNet-based Anti-Spoofing**, which helps identify fraudulent attempts such as printed photographs, mobile screen replays, or static face images. To further strengthen security, the system implements **Active Liveness Verification**, requiring users to perform simple actions such as blinking, ensuring that the authentication process involves a live human presence rather than a spoofed image.

Once liveness is successfully verified, the facial image is processed through **MobileFaceNet**, a highly efficient deep learning model optimized for mobile devices. MobileFaceNet converts facial images into compact high-dimensional facial embeddings, which are then compared against locally stored employee embeddings using **Euclidean Distance Similarity Matching**. This approach enables fast and accurate identity verification without requiring any cloud-based processing. All employee facial templates are securely stored as encrypted embeddings rather than raw facial images, significantly enhancing privacy and reducing storage requirements.

To meet the hackathon requirement of maintaining an extremely lightweight AI footprint, the complete machine learning pipeline undergoes **model optimization and INT8 quantization**, reducing model size while preserving inference accuracy. The optimized models are exported as TensorFlow Lite (.tflite) files, achieving a combined footprint of approximately **20 MB or less**, making them ideal for mobile edge deployment. The optimized inference pipeline enables complete facial recognition and liveness verification in **less than one second**, even on devices with 3GB RAM and no dedicated GPU.

The mobile application is developed using **React Native SDK 54**, enabling a single codebase deployment for both Android and iOS platforms. Real-time camera processing is powered by **react-native-vision-camera**, while optimized on-device AI inference is performed using **react-native-fast-tflite**, ensuring low latency and efficient resource utilization. The application continuously evaluates environmental and facial quality conditions before authentication by providing intelligent guidance such as:

- Face Detected
- Align Face
- Move Closer
- Low Light Warning

These quality checks improve recognition accuracy under challenging field conditions such as harsh sunlight, shadows, and low-light environments.

For offline operation, the application utilizes a local **SQLite database** consisting of two primary tables. The **Employee Table** stores employee information and encrypted facial embeddings, while the **Attendance Records Table** stores authentication events, timestamps, confidence scores, liveness status, and synchronization states. This architecture allows attendance verification to continue uninterrupted in areas with no network coverage.

To support enterprise-scale deployment and future integration with NHAI infrastructure, the solution incorporates an **Offline-to-Online Sync & Purge Mechanism**. Authentication records are stored locally during offline operation and automatically synchronized with AWS cloud services once network connectivity is restored. After successful synchronization, local attendance records are purged, reducing storage overhead and ensuring data consistency across systems.

A key innovation of the proposed solution is its modular architecture. The entire AI authentication engine is designed as a reusable plugin-based SDK that can be seamlessly integrated into existing applications, particularly **Datalake 3.0**, without major architectural modifications. This plug-and-play design significantly simplifies deployment and future scalability across multiple NHAI operational platforms.

By combining Edge AI, Mobile Deep Learning, Facial Biometrics, Anti-Spoofing Security, Model Compression, Offline Data Synchronization, and Cross-Platform Mobile Development, NHAI SecureID delivers a practical, scalable, and government-ready authentication solution. The proposed system achieves the hackathon objectives of lightweight deployment, offline functionality, high accuracy (>95%), sub-second authentication, and seamless Datalake 3.0 integration while providing a secure and future-proof digital identity verification framework for India's highway infrastructure workforce.